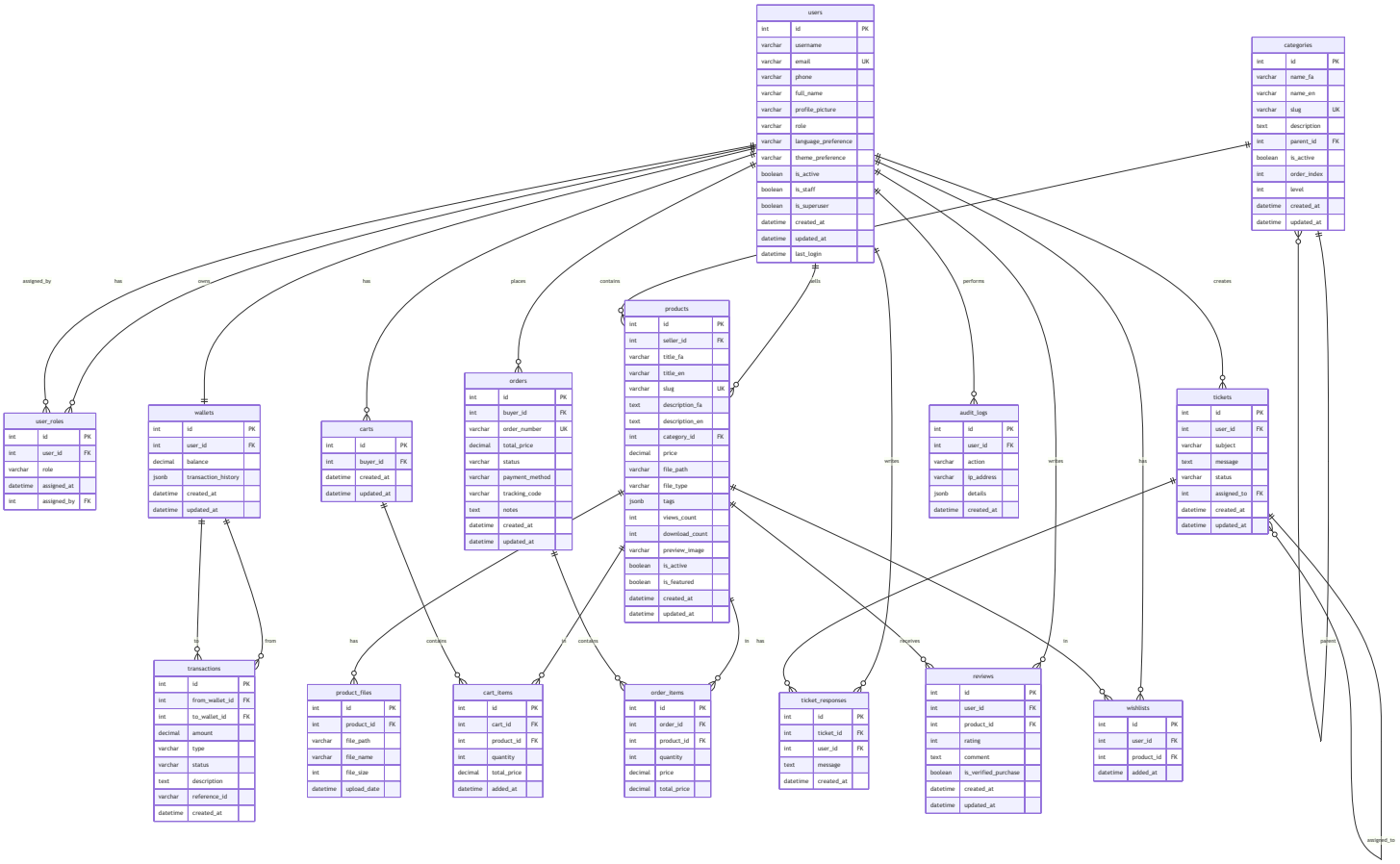
ERD Diagram - Sci Market System

Entity Relationship Diagram



Key Relationships

1. User Management

- **users** → **user_roles**: One-to-many (RBAC system)
- **users** → **wallets**: One-to-one (each user has one wallet)

2. Financial System

- **wallets** → **transactions**: One-to-many (from/to relationships)
- **transactions** can be between different wallets

3. Product Catalog

- **categories** → **categories**: Self-referencing (hierarchical categories)
- **categories** → **products**: One-to-many
- **products** → **product_files**: One-to-many (additional files)

4. Shopping System

- **users** → **carts**: One-to-one
- **carts** → **cart_items**: One-to-many
- **cart_items** → **products**: Many-to-one

5. Order Management

- **users** → **orders**: One-to-many
- **orders** → **order_items**: One-to-many
- **order_items** → **products**: Many-to-one

6. Support System

- **users** → **tickets**: One-to-many
- **tickets** → **ticket_responses**: One-to-many
- **tickets** → **users**: Many-to-one (assigned_to)

7. Review System

- **users** → **reviews**: One-to-many
- **products** → **reviews**: One-to-many
- **reviews** have unique constraint on (user_id, product_id)

8. Wishlist System

- **users** → **wishlists**: One-to-many
- **products** → **wishlists**: One-to-many
- **wishlists** have unique constraint on (user_id, product_id)

9. Audit System

- **users** → **audit_logs**: One-to-many
- **audit_logs** track all user actions

Database Constraints

Primary Keys

- All tables have `id` as primary key
- Auto-incrementing integers

Foreign Keys

- All foreign keys have proper referential integrity
- `ON DELETE CASCADE` for dependent records
- `ON DELETE SET NULL` for optional references

Unique Constraints

- `users.email` - unique email addresses
- `categories.slug` - unique category slugs
- `products.slug` - unique product slugs
- `orders.order_number` - unique order numbers
- `wishlists(user_id, product_id)` - unique wishlist entries
- `reviews(user_id, product_id)` - unique review per user per product

Check Constraints

- `price >= 0` for products and order items
- `quantity > 0` for cart items and order items
- `file_size > 0` for product files
- `balance >= 0` for wallets
- `amount > 0` for transactions
- `rating BETWEEN 1 AND 5` for reviews

Indexes

Performance Indexes

- Foreign key indexes on all FK columns
- Unique indexes on slug fields
- Composite indexes on frequently queried combinations

- GIN indexes on JSONB fields (tags, transaction_history)
- Full-text search indexes on product titles and descriptions

Audit Indexes

- Time-based indexes on created_at fields
- Action-based indexes on audit_logs.action
- User-based indexes on user_id fields

MongoDB Collections (NoSQL)

Event Logs

- High-volume application events
- TTL: 90 days
- Indexes: action, userId, createdAt

Search Analytics

- User search queries and results
- Indexes: query, userId, timestamp

Download Audit

- Product download tracking
- Indexes: productId, userId, timestamp

Ticket Messages (Optional)

- Large ticket conversations
- Indexes: ticketId, userId, timestamp

Summary

This ERD represents a complete e-commerce marketplace system with:

- **16 PostgreSQL tables** with proper relationships
- **RBAC system** for user management
- **Financial system** with wallets and transactions

- **Product catalog** with hierarchical categories
- **Shopping cart** and order management
- **Support system** with tickets and responses
- **Review system** with ratings and comments
- **Wishlist system** for user favorites
- **Audit logging** for all actions
- **MongoDB collections** for analytics and logs

The system is designed for scalability, performance, and data integrity with proper constraints, indexes, and relationships.